

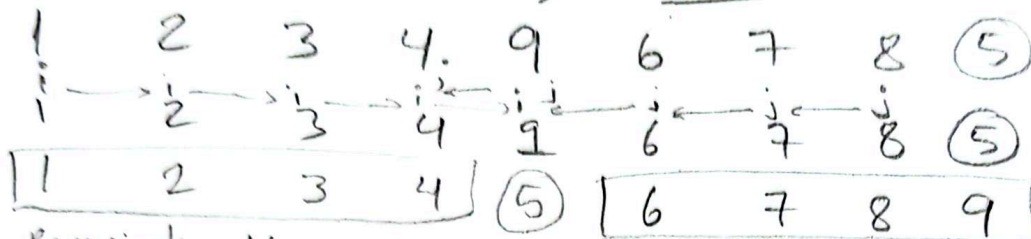
1 2 3 4 5 6 7 8 9

first $A[0] = 1$

last $A[8] = 9$

middle $A[(0+8)/2] = A[4] = 5$

Median of the three $\{1, 9, 5\} \Rightarrow \text{Pivot} = 5$



Recursively call L and R

first $A[0] = 1$

last $A[3] = 4$

middle $A[(0+3)/2] = A[1] = 2$

Median of the three $\{1, 4, 2\}$

Pivot = 2

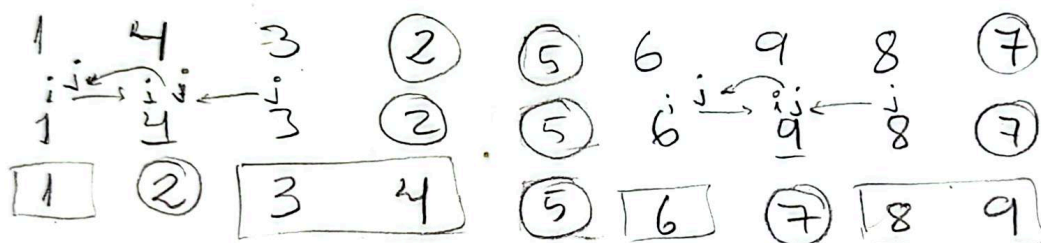
first $A[0] = 6$

last $A[3] = 9$

middle $A[(0+3)/2] = A[1] = 7$

Median of the three $\{6, 9, 7\}$

Pivot = 7



Recursively call L and R branches



first $A[0] = 3$

last $A[1] = 4$

middle $A[(0+1)/2] = A[0] = 3$

Median of the three $\{3, 4, 3\}$

Pivot = 3

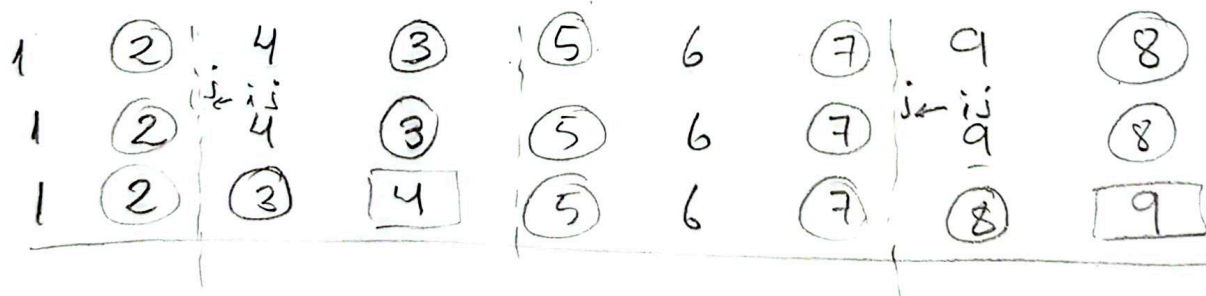
first $A[0] = 8$

last $A[1] = 9$

middle $A[(0+1)/2] = A[0] = 8$

Median of the three $\{8, 9, 8\}$

Pivot = 8



8 7 6 5 4 3 2 1 9

first $A[0] = 8$

last $A[8] = 9$

middle value $A[(0+8)/2] = A[4] = 4$

Median of the three $\{8, 9, 4\} \Rightarrow \text{Pivot} = 8$

9	7	6	5	4	3	2	1	(8)
i							j	
9	7	6	5	4	3	2	1	(8)
1	i	j	i	j	i	j	i	(8)
1	7	6	5	4	3	2	9	(8)
1	7	6	5	4	3	2	(8)	9

Recurisively call L

first $A[0] = 1$

last $A[6] = 2$

middle $A[(0+6)/2] = A[3] = 5$

Median of the three $\{1, 2, 5\} \Rightarrow \text{Pivot} = 2$

1	i	7	6	5	4	3	(2)	(8)	9
j									
1	i	7	6	5	4	3	(2)	(8)	9
1	7	6	5	4	3	7	(8)	9	

first $A[0] = 6$

last $A[4] = 7$

middle $A[(0+4)/2] = A[2] = 4$

Median of the three $\{6, 7, 4\} \Rightarrow \text{Pivot} = 6$

1	(2)	7	5	4	3	(6)	(8)	9
1	(2)	i			j			
1	(2)	7	5	4	3	(6)	(8)	9
1	(2)	3	5	4	7	(6)	(8)	9
1	(2)	3	4	i	j	7	(8)	9
1	(2)	3	4	5	7	(6)	(8)	9
1	(2)	3	4	5	(6)	7	(8)	9

first $A[0] = 3$

last $A[5] = 5$

middle $A[(0+5)/2] = A[2] = 4$

Median of three $\{3, 5, 4\} \Rightarrow \text{Pivot} = 4$

1	(2)	3	5	(4)	(6)	7	(8)	9
1	(2)	i		j				
1	(2)	3	5	(4)	(6)	7	(8)	9
1	(2)	3	4	5	(6)	7	(8)	9

9 1 8 2 7 3 6 4 5

first $A[0] = 9$

last $A[8] = 5$

Middle value $A[(0+8)/2] = A[4] = 7$

Median of the three $\{9, 5, 7\} \Rightarrow \text{Pivot} = 7$

9	1	8	2	5	3	6	4	(7)
i							j	
9	1	8	2	5	3	6	4	(7)
4	→ i	→ i	8	2	5	3	6	j → 9 (7)
4	1	6	→ i	2	→ i	5	→ i	3 → i 8 9 (7)
[4 1 6 2 5 3]					(7)	[9 8]		

first $A[0] = 4$

last $A[5] = 3$

middle $A[(0+5)/2] = A[2] = 6$

Median of the three $\{4, 3, 6\} \Rightarrow \text{Pivot} = 4$

3	1	6	2	5	(4)	(7)
i	→ i	→ i	6	→ i	5	
3	1	6	2	5	(4)	(7)
3	1	2	→ i	6	5	(4)
[3 1 2]			(4)	[5 6]	(7)	(7)

first $A[0] = 9$

last $A[1] = 8$

middle $A[(0+1)/2] = A[0] = 9$

Median of the three $\{9, 8, 9\} \Rightarrow \text{Pivot} = 9$

8	(9)
i	→ i
8	(9)
[8]	(9)
8	(9)

first $A[0] = 3$

last $A[2] = 2$

middle $A[(0+2)/2] = A[1] = 1$

Median of three $\{3, 2, 1\}$

Pivot = 2

3	1	(2)
i	→ i	
3	1	(2)
1	→ i	3 (2)
[1 ✓]	(2)	[3 ✓]

first $A[0] = 5$

last $A[1] = 6$

middle $A[(0+1)/2] = A[0] = 5$

Median of three $\{5, 6, 5\}$

Pivot = 5

(4)	6	(5)
i	→ i	
(4)	6	(5)
(4)	(5)	[6 ✓]
(4)	(5)	6

(7)	8	(9)
(7)	8	(9)
(7)	8	(9)
(7)	8	(9)

